



SMART PORTFOLIO ALLOCATOR: Reinforcement + Generative AI for Financial Strategy Optimization

Capstone Project – University of Arizona (Information Science – ML Track)



TEAM & ROLES

- Shubham Nakhod** – Generative AI Explainability, LLM Integration, RAG Pipeline, Streamlit Dashboard
- Subhrajeet Ghosh** – Reinforcement Learning (PPO/DQN), Environment Design, Policy Optimization
- Jasraj Singh Khalsa** – Data Engineering, Feature Pipelines, Market Dataset Preparation

Mentor: Dr. Bikram Ghosh

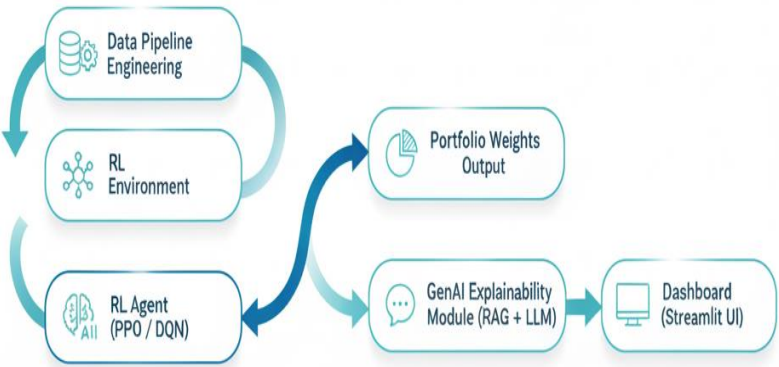
PROJECT MOTIVATION

Financial systems need adaptive, explainable portfolio allocation. Traditional models are static. RL can adapt to market regimes but is a black box. This project combines Reinforcement Learning + GenAI explainability to provide transparent, dynamic investment decisions.

OBJECTIVES

- Build custom multi-asset environment
- Train RL agents (PPO / DQN)
- Integrate GenAI to explain RL decisions
- Construct RAG pipeline for grounding explanations
- Evaluate performance via Sharpe ratio, returns, and drawdown
- Develop explainability dashboard (Streamlit)

SYSTEM ARCHITECTURE



METHODS

Reinforcement Learning:

- Algorithms: PPO, DQN
- Reward: risk-adjusted returns (Sharpe)
- States: asset prices, indicators, portfolio weights
- Environment style: Gymnasium / custom constraints

Generative AI Module:

- LLM-based explanations of strategy
- RAG using embeddings (FAISS/Pinecone)
- Converts RL outputs into natural language insights

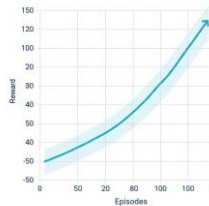
Data Engineering:

- Market data: Yahoo Finance / Kaggle
- Features: returns, volatility, technical indicators
- Automated ETL pipelines

VISUALIZATIONS

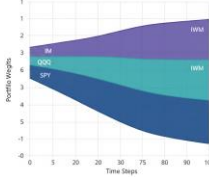
1. RL Learning Curve (PPO/DQN)

RL Learning Curve (PPO)



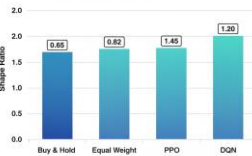
2. Portfolio Allocation Trajectory

Portfolio Allocation Trajectory



3. Sharpe Ratio Comparison

Sharpe Ratio Comparison



EXPECTED OUTCOMES

- Working RL portfolio allocator
- Fully explainable decisions via GenAI
- Integrated financial AI dashboard
- Improvement over buy-and-hold & equal-weight strategies
- Transparent AI suitable for real-world financial workflows